

ESEMPIO

$$\bullet \int \frac{1}{x(x^2+2x+2)} dx = ?$$

Sia

$$f(x) = \frac{1}{x(x^2+2x+2)} = \frac{A}{x} + \frac{Bx+C}{x^2+2x+2}$$

allora

$$A(x^2+2x+2) + x(Bx+C) = 1 \iff \begin{cases} A+B=0 \\ 2A+C=0 \\ 2A=1 \end{cases} \quad \text{da cui } A = \frac{1}{2}, \quad B = -\frac{1}{2}, \quad C = -1$$

così

$$\int \frac{1}{x(x^2+2x+2)} dx = \int \left(\frac{1/2}{x} - \frac{1/2 x + 1}{x^2+2x+2} \right) dx = \frac{1}{2} \log|x| - \int \frac{1/2 (\ell-1) + 1}{(\ell-1)^2 + 2(\ell-1) + 2} d\ell = \frac{1}{2} \log|x| - \frac{1}{2} \int \frac{\ell+1}{\ell^2+1} d\ell = \frac{1}{2} \log|x| - \frac{1}{2} \int \frac{\ell}{\ell^2+1} d\ell - \frac{1}{2} \int \frac{1}{\ell^2+1} d\ell$$

$x = \ell - 1$

$$= \frac{1}{2} \log|x| - \frac{1}{4} \int \frac{1}{\ell^2+1} d(\ell^2+1) - \frac{1}{2} \int \frac{1}{\ell^2+1} d\ell = \frac{1}{2} \log|x| - \frac{1}{4} \log(\ell^2+1) - \frac{1}{2} \arctan(\ell) + c = \frac{1}{2} \log|x| - \frac{1}{4} \log(x^2+2x+2) - \frac{1}{2} \arctan(x+1) + c$$

$$\bullet \int \frac{1}{\sin(x)} dx = \int \frac{\sin(x)}{1-\cos^2(x)} dx = \int \frac{-d\ell}{1-\ell^2} = \int \left(\frac{A}{\ell-1} + \frac{B}{\ell+1} \right) d\ell = \frac{1}{2} \log|\ell-1| - \frac{1}{2} \log|\ell+1| + c = \frac{1}{2} \log\left(\frac{1-\cos(x)}{1+\cos(x)}\right) + c$$

$\ell = \cos(x)$
 $d\ell = -\sin(x) dx$

$$\bullet \int \frac{1}{x \log^2(x) (\log^2(x)+1)} dx$$

$$= \int \frac{1}{\ell^2 (\ell^2+1)} d\ell = \int \frac{A}{\ell} + \frac{B}{\ell^2} + \frac{C\ell+D}{\ell^2+1} d\ell = -\frac{1}{\ell} - \arctan(\ell) + c = -\frac{1}{\log(x)} - \arctan(\log(x)) + c$$

$\ell = \log(x)$
 $d\ell = \frac{1}{x} dx$

$$\bullet \int \frac{e^{2x} + 2e^{-x}}{1 + e^x} dx$$

$$= \int \frac{\ell^2 + 2\ell^{-1}}{1 + \ell^{-1}} \cdot \frac{d\ell}{\ell} = \int \frac{\ell^3 + 2}{\ell(\ell+1)} d\ell = \int \left(\ell - 1 + \frac{A}{\ell} + \frac{B}{\ell+1} \right) d\ell = \frac{\ell^2}{2} - \ell + 2 \log|\ell| - \log|\ell+1| + c = \frac{e^{2x}}{2} - e^x + 2x - \log(e^x+1) + c$$

$\ell = e^x$
 $x = \log(\ell)$
 $dx = \frac{d\ell}{\ell}$

$$\bullet \int \frac{1}{\sqrt{1+e^x}} dx = \int \frac{1}{\sqrt{1+\ell}} \frac{d\ell}{\ell} = \int \frac{1}{s} \cdot \frac{2s ds}{s^2-1} = \int \frac{1}{s-1} - \frac{1}{s+1} ds = \log|s-1| - \log|s+1| + c = \log\left(\frac{s-1}{s+1}\right) + c = \log\left(\frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1}\right) + c$$

$\ell = e^x$
 $dx = \frac{d\ell}{\ell}$
 $s = \sqrt{1+\ell}$
 $s^2 = 1+\ell$
 $2s ds = d\ell$

OSSERVAZIONE

Se g è continua in $[a, b]$ e f è continua in $g([a, b])$ allora:

$$\int_a^b f(g(x)) g'(x) dx = \int_{g(a)}^{g(b)} f(t) dt$$

$t = g(x)$
 $\downarrow$
 $g(a)$ $g(b)$

ESEMPI

$$\int_0^{\pi/4} \sin^2(x) dx = \frac{1}{2} \int_0^{\pi/4} (1 - \cos(2x)) dx = \frac{1}{2} \int_0^{\pi/4} (1 - \cos(t)) \frac{dt}{2} = \frac{1}{4} [t - \sin(t)]_0^{\pi/4} = \frac{\pi}{8} - \frac{1}{4}$$

$t = 2x$
 $dt = 2dx$

Perché $\cos(2x) = \cos^2(x) - \sin^2(x) = 1 - 2\sin^2(x)$ implica $\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$

$$\int_0^8 \frac{1}{1+\sqrt{x}} dx = 3 \int_0^2 \left(t - 1 + \frac{1}{1+t} \right) dt = 3 \left[\frac{t^2}{2} - t + \log|1+t| \right]_0^2 = 3 \log(3)$$

$t = \sqrt{x}$
 $t^3 = x, 3t^2 dt = dx$

$$\int_{\pi/2}^{\pi} \sin(2x) \log(1 - \cos(x)) dx = -2 \int_0^1 t \log(1-t) dt = \int_{-1}^0 \log(1-t) d(t^2) = \left[t^2 \log(1-t) \right]_{-1}^0 - \int_{-1}^0 -\frac{t^2}{1-t} dt = -\log(2) - \int_{-1}^0 \left(t+1 + \frac{1}{t-1} \right) dt$$

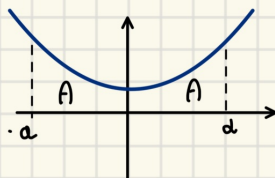
$t = \cos(x), dt = -\sin(x) dx$

$$= -\log(2) - \left[\frac{t^2}{2} + t + \log|t-1| \right]_{-1}^0 = -\log(2) + \frac{1}{2} - 1 + \log(2) = -\frac{1}{2}$$

OSSE RVAZIONE

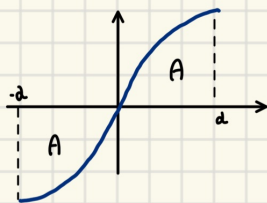
Se f è pari in $[-a, a]$ allora

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$



Se f è dispari in $[-a, a]$ allora

$$\int_{-a}^a f(x) dx = 0$$



$$\bullet \int_{-1}^1 e^{-x^2} (|x| + \sin(x)) dx = \int_{-1}^1 e^{-x^2} |x| dx + \int_{-1}^1 e^{-x^2} \sin(x) dx = 2 \int_0^1 e^{-x^2} |x| dx + 0 = [-e^{-x^2}]_0^1 = 1 - e^{-1}$$

$$\bullet \int_{-3}^3 \lfloor x \rfloor dx = (1+2) - (1+2+3) = -3$$